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KE1186

March 22, 2021

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Changing Banking's DNA:

The Impact of Fintechs on Small Business Lending

The rise of financial technology companies—"fintechs"—is changing the way money moves around the world, leading to greater financial inclusion and closing a credit gap that historically has hampered small businesses. According to the World Bank, small and medium enterprises (SMEs) is the engine driving world economies, representing 90 percent of all businesses worldwide, providing 50 percent of all employment, and responsible for up to 40 percent of national income in developing economies.[1] Even so, research has shown that 65 million businesses—or 40 percent of SMEs in developing economies—face an unmet financing need of $5.2 trillion per year.[2] This gap translates to 19 percent of the gross domestic product of the 128 countries surveyed[3] and attests to the vital role that SMEs play in the world economy as a driver of employment and overall economic health.

Fintech and tech companies such as Alibaba, Amazon, and Square have stepped into this gap as their payment platforms have evolved into gateways for expanded financial service offerings. Although the companies have pursued different strategies in their quest to fill e-commerce needs, their innovation shares a common focus on leveraging data, networks, and artificial intelligence to expand financial services, evaluate creditworthiness, and ultimately enable individuals and businesses to secure funding that propels growth. Many alternative lenders to date have targeted those whom the conventional banking system largely ignored due to risk and expense, but a banking showdown between digital empires and legacy financial institutions seems increasingly plausible. Although most fintech companies do not have the banking licenses that would lead to direct competition with conventional banks, their increasing interest in financial services—everything from payments to loans to wealth management to insurance—has revolutionized lending and carries lasting implications for the rest of the financial services industry.

Banks and the Economics of Small Business Lending

Of the vast numbers of small and medium businesses that fail each year, nearly half of entrepreneurs blame lack of funding or working capital.[4] Small companies often face challenges in obtaining financing to bring a new product to market, fund expansion, or pay for ongoing marketing costs. Without cash, many small business owners are forced to close their doors.

Traditionally, banks preferred collateral-based lending and required land, equipment, or goods as a guarantee in exchange for financing. To be more inclusive, conventional banking gradually moved toward a risk-based system that incorporated credit histories and financial information to determine creditworthiness. Even this shift did not help many SMEs, though, as they tended to have less-stable sources of revenue, limited liquidity, and insufficient data for credit risk assessment, thereby leading to higher interest rates or outright rejection of financing. However, the increasing availability of online data began changing the landscape in the early 2000s when new data sources from social and economic activity became available to help fill the credit scoring gap, and information-based lending began to take shape.

The traditional banking system is one of the largest sectors in the world, but until recently, it was also one of the least competitive due to high barriers of entry. In China, the United Kingdom, India, and France, the five largest banks in each country dominated more than half of their respective markets.[5] Most financial institutions are very capital intensive because they require extensive networks of physical buildings even in the most rural communities, as well as educated staff, ATM networks, and technology. Registration costs alone to set up a bank in the United States can run to millions of dollars.

Conventional banks also possess core data assets. Lending requires large banks to have verified and reliable information, with a long history of both “soft” data from loan officers’ personal interactions with customers and “hard” data related to business performance, account, and financial histories; in addition, a legacy of data privacy protection encourages customer trust. However, banks typically have a more limited range of non-financial activities from which to collect data than their big tech counterparts do. Legacy technology sometimes limited their data processing capabilities, and transactional data was sometimes one-sided.

Despite these challenges, traditional banking institutions benefit from their depth of experience in risk management and relatively easy access to large and inexpensive funding sources. For example, China’s top six banks had deposits of more than US$13 trillion in 2019.[6] In contrast, fintechs often struggle with higher costs of capital.

An Overview of SME Lending in China

Historically, China's banking industry was dominated by large, state-owned banks that paid low interest on consumer deposits and lent money primarily to local governments and to large, state-owned enterprises. Banks favored lending to state-owned enterprises rather than to the private sector because lending to SMEs appeared riskier and more time-consuming due to factors including a lack of collateral, credit histories, and transparent financial records. This trend was particularly prevalent after the global economic meltdown in 2008. China's stimulus package to build infrastructure and jobs led to the largest credit expansion of any single country in history, as the banking system grew from $9 trillion in 2008 to $38 trillion in 2017.[7] Despite increasing evidence that state-owned enterprises are less efficient than their private counterparts regarding resource allocation, productivity, and long-term growth,[8] the trend toward funding the public—rather than the private—sector has become more pronounced in recent years. In 2014, 60 percent went to the state sector, and 34 percent went to the private sector; but in 2015, 69 percent went to the state sector, and 19 percent went to the private sector; and in 2016, 83 percent went to the state sector, and only 11 percent went to private companies.[9]

China's state department reported that the country had about 73 million SMEs in 2017, and a 2018 joint report between the People's Bank of China and the World Bank Organization revealed that only 14 percent of SMEs had access to credit.[10] The report cited a variety of barriers, including complex application processes, unfavorable interest rates, insufficient loan size, and long review and approval periods. To further complicate matters, the use of traditional credit cards is not widespread in China; only about one-fifth of China's population held them in 2017.[11] Most individuals in need of capital were left to pursue assistance from friends, family, or informal, unregulated shadow lending entities.[12] As a result, most SMEs lacked affordable and reliable access to the capital necessary to grow their business or to withstand an economic downturn.

During the past few years, the Chinese government has prioritized SME growth, prompting important changes in the policy environment. The Law of the People's Republic of China on the Promotion of Small and Medium-Enterprises (2017), known as the SME Promotion Law, came into effect on January 1, 2018, and sought to strengthen public and private financing for SME development and to promote bond market and equity financing channels. In addition, it strove to provide tax benefits, to streamline procedures for establishing and listing businesses, to reduce administrative costs, and to facilitate foreign exchange and the development of international SMEs.[13] A 2019 government campaign urged banks to boost lending to SMEs by more than 30 percent and increased the tolerance for non-performing loans to small companies.[14] Despite concerns about the economy's sluggish growth amid roiling trade tensions with the United States, evidence of positive transformation is emerging: Chinese banks lent an unprecedented Rmb 2 trillion ($286 billion) to small companies with limited access to credit in the first ten months of 2019 (up from Rmb 1.7 trillion for all of 2018). The Industrial and Commercial Bank of China, the nation's largest lender by assets, increased small business lending by 48 percent in the first 11 months.[15] Clearly, progress is being made, although some alarms are being raised as well: Many banks are lending at less-profitable interest rates, and the rate of non-performing small business loans reached 5.9 percent in May 2019, as compared with 1.4 percent on loans to larger borrowers.[16]

An Overview of SME Lending in the United States

After the financial crisis in 2008, banks sharply reduced small business lending, creating a debilitating credit supply gap. Between 2008 and 2011, small business lending fell $116 billion—almost 18 percent—from $659 billion to $543 billion, and small commercial and industrial lending dropped more than 20 percent during the same period.[17] Small business lending by the four largest U.S. banks fell sharply relative to other banks in 2008, and this decline was particularly significant because almost three-quarters of small businesses considered major banks to be their primary financial institution.[18] Even as the macroeconomic situation began to improve in subsequent years and small businesses began forecasting growth and profits, the credit gap remained, constraining small businesses and resulting in a sluggish recovery for the wider U.S. economy. In 2014, small business loan originations for the top four banks were still only at half their 2006 levels, whereas other banks' originations had recovered up to 80 percent of 2006 levels.[19]

Unfortunately for SMEs, the high costs and volume constraints associated with conventional business underwriting and review processes caused many institutions to abandon the lower dollar loan amounts that SMEs typically sought in favor of higher balance credit offerings that could be extended more profitably.[20] Studies demonstrated that a traditional underwriting process entailed nearly 30 separate tasks involving multiple departments at a bank to arrive at a lending decision. These tasks included collecting information and financial statements for the credit application, reviewing the financial information, completing data entry and calculations, performing industry analytics, assessing borrower capability and capacity, and evaluating collateral. A manual underwriting process could consume up to 100 hours, cost approximately $3,000 to 5,000 per application, and take 30 days or longer to complete (see **Exhibit 1**).[21] Moreover, the costs associated with a $100,000 loan are roughly the same as those for a $1 million loan.[22] Given these parameters, a typical institution could handle ten loan applications per analyst per month. At an average booking rate* of 50 percent, this correlated to an average underwriting cost of $10,000 per loan.[23] In addition, a loan review process typically involved 15 steps over two days, at the cost of $1,000 per loan, involving a bank's lending, credit, and audit departments (see **Exhibit 2**).[24]

The implications of the inefficiencies endemic to these heavily manual underwriting and review processes are more clearly underscored by reviewing the earnings implications for different loan amounts. Ninety-five percent of potential small business borrowers had loan needs of less than US$250,000, and the average SME loan amount was $75,000.[25] However, as seen in **Exhibit 3**, the break-even loan amount was $861,505 for a 60-month term loan charging 5 percent interest with a desired return on equity of 15 percent and a 10 percent capital requirement.[26] Furthermore, a similar $75,000 loan delivered an ROE of -30.89 percent.[27] In addition, an institution with a thousand commercial contracts would sustain more than $1 million in costs to conduct annual loan reviews— and likely even greater costs given that some loans with riskier profiles required reviews between two and four times per year instead of just once. High costs and substantial time and resource requirements caused lenders to move upmarket in search of higher value loans, thereby hampering access to credit and creating a shortage of loan offerings with favorable terms and approval processes for small businesses.

* The booking rate is the number of loans actually transacted in relation to the number of applications received.

However, the growth of fintech companies, as well as partnerships between fintechs and traditional institutions, began changing the lending landscape for SMEs. Fintech companies revolutionized the loan application experience for entrepreneurs in need of capital. Online applications skyrocketed, and advanced data analysis made lending decisions available at record speeds. Instead of compiling voluminous paperwork and waiting weeks for a decision, small business owners had the flexibility to apply for financing online at their convenience rather than during business hours at a bank branch, and lending decisions could be made in days or even minutes rather than weeks or a month. Many banks began partnering with companies that could boost their digital lending capabilities, optimizing outcomes for lenders, borrowers, fintech providers, and wider consumers and local economies alike.

By 2017, around 48 percent of small businesses were expressing interest in applying for credit (up from just 19 percent in 2012) as they sought to capitalize on perceived optimism in market conditions.[28] Applications to alternative lenders continued to grow, with 32 percent of applicants turning to online providers in 2018.[29] However, the Small Business Credit Survey revealed that 53 percent of small businesses did not receive the amount of financing they sought in 2019. Loan approval rates were 58 percent at large banks, 71 percent at small banks, and 82 percent at online lenders.[30]

The Rise of Fintech

China is at the forefront of fintech growth and is the largest fintech market in the world, drawing nearly half of the $55 billion global fintech investments in 2018.[31] An estimated 890 million unique individuals make transactions valued around $17 trillion a year in China, securing the country's position as the world's leading mobile payments market.[32] Ant Financial's Alipay controlled more than half of the Chinese payment market, and the company, together with its competitor Tencent, handled 94 percent of all mobile transactions in China.[33] Some speculate that the sheer inconvenience of former cash-based systems has helped prompt the mobile payment revolution and its accompanying financial services expansion in developing countries. This revolution in turn has built vast networks of data that can be leveraged for everything from wealth management to evaluating credit risk. This is especially valuable in China, where the number of unbanked is large and there is no central credit scoring system.

In contrast, the United States has seen much slower adoption rates of mobile payments despite the near-ubiquity of smartphones. Card usage (both credit cards and debit cards) continues to dominate the noncash payment space, accounting for 131.2 billion transactions valued at more than $7 trillion in 2018.[34] Many U.S. consumers are content to use their existing credit and debit cards and express concern about privacy and mobile security. Despite consumers' reluctance to adopt mobile payments, digital transformation of the financial services industry in the United States has been significant, as well.

Big Tech's Foray into Finance

Technology firms such as Alibaba, Amazon, and Square have experienced explosive growth during the past twenty years. Although all of these companies began with a practical solution to a specific payment problem, their business models facilitated direct interactions among a large number of users, thereby creating vast amounts of data that could be leveraged to offer a variety of services that further exploited network effects. These in turn generated additional user activity and data that perpetuated an ever-widening "data-network-activities loop."[35] Given the economic power and opportunity that sheer size and scale impart, all three of these companies or their related entities have ventured into financial services such as lending, wealth management, and insurance.

Alibaba and Ant Financial

In 1999, a group of 18 people led by a former English teacher named Jack Ma established Alibaba as a website to help Chinese small exporters, manufacturers, and entrepreneurs sell internationally. Twenty years later, Alibaba has grown into the world's largest online and mobile e-commerce company, with a market capitalization of $570 billion.[36] The company's meteoric rise reflects its commitment to innovation and to expanding financial inclusion to bring greater value for society and equal opportunities to the world.

Alibaba began as the business-to-business platform Alibaba.com. In 2003, the company expanded to include a consumer-to-consumer marketplace called Taobao.com, similar to that of eBay in the United States. However, sales were sluggish due to trust issues between buyers and sellers. As a result, to alleviate fears around online shopping, Alibaba in 2004 established Alipay, a third-party online payment platform similar to PayPal. Acting as an internet escrow service, Alipay received customer payments but guaranteed that no money would be transferred until items or services were delivered as promised. The following year, Alipay partnered with the Industrial and Commercial Bank of China to deliver its online payment services, and the company gradually began offering offline payment services as well. The first offline use enabled Alipay users to pay their utility bills via the online platform instead of in person. The safety, convenience, and efficiency of digital payments earned people's business, and both digital payments and e-commerce sales in China took off. Alipay became the world's leading digital lifestyle and payment platform, serving more than 1.2 billion users, along with its global e-wallet partners, in 2019.[37] Through an open-platform strategy, Alipay connects consumers with local, public, and financial service providers, and it helps merchants digitalize their operations, thereby promoting digital transformation in the service industry.

From the company's inception, Alibaba aggregated insights such as customer purchasing habits, transactions, and comments on merchant sites, providing and analyzing valuable marketing, financial, purchasing, merchandising, and inventory information for merchants on its platform. Alipay was spun off in 2010 as a separate affiliated entity, although it continued to give Alibaba 37.5 percent of pre-tax profits each quarter.[38] That same year, Alibaba also began experimenting with lending to merchants on its platforms through Ant Micro Loan. Whereas the minimum loan size at a conventional bank at the time was around Rmb 6 million (US$1 million), Ant Micro

Loan had a maximum loan of Rmb 1 million (US$160,000) when it started.[39] The smaller size more closely matched the needs of SMEs, and the loans enabled sellers to expand their businesses, thereby generating more growth and revenue on Alibaba's platforms and reinforcing the company's core business.

In 2014, Alibaba went public with a $25 billion initial public offering, the largest in history at the time. The following month, the company rebranded Alipay as Ant Financial Services Group. Eventually, the relationship between the companies was restructured, and Alibaba exchanged its profit-sharing agreement for a 33 percent equity stake in Ant.[40] The name "Ant" reflected the company's intention to serve individuals and SMEs and to demonstrate the strength and power that comes from working together. Five years later, Ant had lived up to that sentiment, and the $150 billion company had earned the twin titles of the world's most valuable unicorn and its most valuable fintech company.

Pioneering a Digital Ecosystem

Ant Financial envisioned itself not merely as a marketplace or a payment provider, but as a "lifestyle platform" or "ecosystem" that meets a host of consumer needs. Building on success in the payments sphere, Ant began layering other products on top of the platform, offering everything from wealth management, lending, and insurance products to those enabling ride-sharing, food ordering, and entertainment purchases.

Payments and aggregate data from transactions continue to be a core business area and asset for Ant. In contrast to credit card companies that routinely charge 2.5 to 3.5 percent transaction fees, Alipay and other Chinese payment platforms tend to reduce or eliminate fees and intermediaries from consumer spending transactions. The proprietary system is built on digital wallets and QR codes rather than on bank-based credit cards or cash, and transactions run primarily through big tech firms. Digital wallets in China typically are linked to a bank account that transmits funds. An Alipay user (whether an individual, a merchant, or a specific payment point such as a parking garage) has a QR code, a unique, two-dimensional bar code that can be digitally represented on a smartphone or physically printed on a piece of paper. When payment is initiated, either the buyer or the seller scans the QR code; the payer can total the amount due, or the payee can insert the amount to be paid. This process transfers money directly between Alipay accounts and eliminates the need for a credit card–reading terminal or any other processing intermediary. Sellers pay a transaction fee of less than 1 percent; however, to encourage participation among micro merchants, the fees are refunded for those whose monthly transaction volume fell below a certain threshold.[41] Buyers do not pay fees for using Alipay. In 2016, Alipay instituted a 0.1 percent transaction fee for users who wished to transfer funds above Rmb 20,000 (US$3,000) from their Alipay accounts back into a bank account. The move incentivized users to keep money in their Alipay wallet, and these readily available funds could in turn lead to higher activity in Ant's ecosystem. In addition to Alipay, Ant oversees a number of other financial service offerings, including MYbank to offer both loans and lines of credit; Ant Fortune to provide wealth management opportunities; a platform for insurance providers; and a suite of technology products and services.

MYbank was established in 2015 to assist SMEs and farmers, particularly in rural areas, who were largely unserved by conventional banks. MYbank was the first bank in China set up entirely on the cloud without physical branches, and it required all accounts to be linked with Alipay accounts, thereby leveraging transaction histories to better understand cash flow. MYbank pioneered the "310" lending model and offered SMEs or farmers business loans (with either online or offline operations) that could be applied for through a mobile phone in less than three minutes. These loans took less than one second to obtain approval, and the process required zero human intervention. Rather than requiring collateral or voluminous amounts of paperwork, MYbank was able to assess creditworthiness using Ant's real-time payment data, artificial intelligence technologies, and risk management systems that analyzed more than 3,000 variables. When loan applicants granted MYbank permission to access their data, algorithms could instantly analyze all information generated inside the Alibaba network.[42] MYbank President Jin Xiaolong reported in 2018 that traditional lenders typically took 30 days to process loans and rejected 80 percent of small business loan requests. In contrast, he stated that MYbank approved 80 percent of loans in a process that took only a few minutes, and because the process was automated, the operating cost per loan was 3 yuan (less than US$1) instead of the 2,000 yuan (US$280) cost at traditional banks.[43]

By the end of 2019, MYbank had provided more than Rmb 2 trillion (US$290 billion) in loans to more than 20 million small and microenterprises and entrepreneurs in China. Although loan sizes could climb to Rmb 1 million (US$141,394), a typical MYbank loan was Rmb 9,900 (US$1,469). The bank was able to support Alibaba's sellers in a targeted way; for example, MYbank loaned Rmb 300 billion to three million sellers in preparation for a Chinese shopping festival known as Singles Day in 2019 that ultimately resulted in the sale of more than US$38.3 billion in merchandise on Alibaba's platforms.[44] Many of MYbank's small business customers became recurring customers, using loans to assist with anything from building inventory to handling short-term cash flow. Loan payments were made through Alipay or the MYbank app, and the non-performing loan ratio hovered around 1 percent.

Ant offers a complete suite of technology products and services covering all the "BASICS" to financial institutions interested in using its digital lending platform: Blockchain, AI, Security, Internet of Things, and Computing. Together these solutions enable the delivery of services efficiently and at scale and generate concrete gains for institutional partners who implement them. For example, the Bank of Nanjing began online consumer finance operations in 2016. Before the online launch, the bank had fewer than 200,000 offline contracts. In one year, the number increased to five million.[45] The bank moved its asset management, insurance, lending, and payments products to the new platform, where it processes more than one million transactions per day. By the end of 2018, loans under management exceeded Rmb 82 billion.[46]

In 2018, Ant Financial announced an initiative to open up a full suite of its technology products to support the growth of financial institutions, including helping them to optimize the user experience and to reduce costs. In addition, MYbank announced a new initiative called the Star Plan and pledged to share its technology services with 1,000 financial institutions to provide financing services to 30 million SMEs and farmers in China in the next three years.[47]

In recent years, Ant further enhanced collaboration with financial institutions. Ant shares technologies and customer engagement tools to help partners as they seek to adapt to the mounting regulatory changes, as well as to meet the needs of consumers looking to make investments, buy insurance, and secure financing. Ant uses Alipay's payment services as a gateway to its financial services ecosystem and, increasingly, to other digital lifestyle services; indeed, most of Ant's users are active in all of its financial service areas. The sheer scale of Ant's vast network of hundreds of millions of users and its ability to leverage data and cross-sell products and services on its platforms make participation virtually irresistible for financial service providers wishing to stay relevant in a digital age. Ant also provides technological solutions to clients, so it increasingly relies on technology and service fees for revenue. The company anticipates generating as much as 65 percent of its revenue from technology services by 2021, up from 34 percent in 2017.[48]

Amazon

In 1994, Jeff Bezos founded Amazon with $10,000 as an online bookstore in his garage. Twenty-five years later, the business has become the most valuable company in the world, with 750,000 employees and a market capitalization of around $1 trillion. Known as the "Everything Store," Amazon has moved into nearly every type of product, media, and service, accounting in 2019 for more than half of all US online retail sales and nearly 14 percent of worldwide retail sales.[49] A crucial byproduct of its business is a massive supply of data on everything from customers and transactions to pricing trends and supply chains.

In recent years, Amazon has made strides in the financial services sector as it seeks to facilitate merchant and customer participation in its ecosystem. Amazon Lending does not check credit reports. Instead, it uses transaction data and other available data, such as the merchant's product return rates, delivery problems, or customer complaints, to get insight into the business's stability and quality. Amazon Lending does not ask for any application, origination, or prepayment fees.[50]

Amazon has built a series of payment options into its platforms. Buyers can make purchases on Amazon using a debit card, an Amazon Prime Store credit card, and a host of other rewards credit cards, points, or cash back from rewards cards, gift cards. They can also use Amazon Pay digital wallet, Amazon Cash accounts, Amazon PayCode, and Amazon Currency Converter.

Amazon Lending

Amazon has 1.9 million small businesses selling in its marketplaces, and its SME sellers generate more than half of all worldwide sales.[51] Amazon Lending, established in 2011, seeks to encourage growth by providing short-term loans to its merchants.

Amazon offers its sellers invitation-only loans to customers with good metrics and profitability who have been selling on Amazon for at least a year. Amazon provides loans in a range of $1,000 to $1 million that last between three and twelve months. The average loan size was in line with fintech industry benchmarks—around $5,500.[52] Interest rates typically range from 6 percent to 17 percent.[53] To apply, sellers accept offers emailed to their seller sites. Applying generally requires only a few clicks, and former borrowers report being approved and receiving the funds as quickly as the same day and not longer than one week.[54] Amazon Lending does not check credit reports and asks for no application, origination, or prepayment fees.[55]

However, loans from Amazon Lending are not appropriate for everyone and must be considered carefully. Limited information is available to borrowers about loan eligibility and interest rates, and sellers cannot solicit offers; they must wait for an invitation. If approved, the funds are used to expand an Amazon business by buying inventory, gaining cost or production efficiency, investing in product development and advertising costs, or building business infrastructure.[56] Amazon deducts payments from the seller account each month. If sellers do not have enough on their due date to cover the balance, the company will deduct the remainder from the next seller disbursement. In the event of non-payment, Amazon has the option of seizing inventory or seller account funds until the balance has been paid.

Despite these potential downsides, Amazon Lending loaned more than $1 billion to SMEs in 2018 in a successful push to boost marketplace growth.[57] Amazon's default rate for SME loans was about 75 percent lower than other fintechs and banks. Whereas OnDeck's default rate was around 8 percent,[58] and Lending Club charged off more than 10 percent of total loans issued,[59] Amazon's default rate was about 2 percent. In February 2020, banking giant Goldman Sachs was in advanced talks with Amazon to offer small business loans through Amazon's lending platform, which would reduce Amazon's lending risks while significantly expanding Amazon sellers' access to capital.[60]

Square

Founded in San Francisco in 2009, Square launched a signature card reader that enabled small businesses to accept credit and debit card payments by connecting a card-reader dongle to mobile devices like smartphones and tablets. Less than a decade later, more than thirty million businesses use its technology, and it has evolved into a $30 billion company with a full suite of financial service products, including Cash App, payroll services, business loans, business analytics, and bitcoin.

Square Cash App

Square launched a peer-to-peer (P2P) payment service in 2013 to compete with rivals such as Apple Pay, Google Pay, and PayPal's Venmo. To set up an account, users create either a business account or a personal account, depending on their intended usage (or they can set up two accounts if they would like to have both types) and link those accounts to a bank account. Business accounts can only receive money, whereas personal accounts can both send and receive money. Business accounts are charged 2.75 percent of each transaction. On the personal account side, Cash App does not charge fees to send money from debit accounts; instead, it charges the user a 3 percent fee to send money via a credit card. Users with Siri on their smartphones can set up voice-activated payment. When users receive cash, the funds can either stay in the Cash App account or users can request a free Cash Card that serves as a debit card for the Cash App account. Users who elect to return Cash App funds to their bank account must wait several days for the funds to become available if they make a standard deposit. Alternatively, they can pay a 1.5 percent transaction fee to transfer the funds via instant deposit so that the funds ca be accessible within thirty minutes.

Unlike its competitors, Square was not content to allow its payment app to remain simply a payment app, and it began building in other functionalities. In 2017, Cash App introduced a bitcoin exchange so users could apply residual funds to the buying and selling of bitcoins. In 2018, Square enabled Cash App to receive direct deposits and introduced “Boosts,” which turned the free Cash Card into a rewards card and provided instant cash back offers from select retailers and restaurants. In 2019, the company launched a free debit card for businesses called Square Card. The Square Card enables sellers to access Cash App funds as soon as they are delivered rather than having to wait additional time for funds to clear in their bank account. The Square Card also offers sellers a 2.75 percent discount on purchases made from other Square businesses.[61] Square launched an in-app stock-trading feature that allows users to buy and sell stocks or even fractions of stocks free of charge. Cash App’s growing ecosystem propelled its number of users from seven million to fifteen million in 2018 as it overtook Venmo to become the nation’s leading P2P app downloaded from the App Store.[62] By 2019, the Square Cash App had been downloaded 59.8 million times since its inception, whereas the Venmo app had been downloaded 52.7 million times.[63] With these impressive gains in customer usage, the Square Cash App is quietly transforming itself into a digital bank by offering basic services such as direct deposits and debit cards and by encouraging people to hold account balances.

Square Capital

In 2014, Square launched Square Capital, which extends business loans ranging from $500 to $250,000 to merchants in its network. Like Amazon Lending, the service is offered by invitation only, and the average loan size is around $6,000.[64] Applying takes only a few minutes, and the process has no intricate forms, credit checks, or waiting periods. Algorithms evaluate data from a host of factors—including account history, payment processing volume, payment frequency, and retailer inventory levels—to determine a business’s eligibility and to make lending decisions.

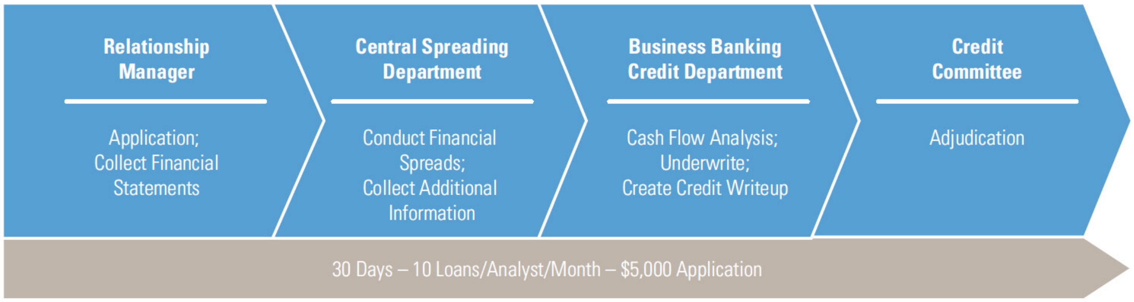
If approved, a business loan is deposited in the seller’s bank account as early as the next business day. Sellers pay a loan fee to borrow the loan but are assessed no other late fees, prepayment fees, or surprises. Sellers agree to pay a fixed percentage of daily card sales that will be automatically deducted until the loan is repaid. If daily card sales do not cover the minimum payment, Square Capital will debit the remaining minimum payment amount due from the seller’s linked bank account. Loans must be paid back within 18 months, and sellers must meet a minimum payment every 60 days. Square Capital has extended $5.5 billion in loans and merchant cash advances to more than 275,000 merchants, and 95 percent of them said they grew their business thanks to the loan.

In May 2018, Square Capital expanded its offerings to include consumer installment loans for purchases of $250 to $10,000 at Square merchants in 22 states. Although the move complements the company’s payment processing business, it also takes on risk. Although it is growing its consumer data through Cash App, Square historically has been focused on tracking merchant data. However, Square’s CFO said it was the next step in the company’s move into more traditional banking and noted that shareholders should expect continued activity in that direction in the future.[65] The company applied for a banking charter in 2017, withdrew the application briefly to engage in discussions with regulators, and then reapplied in 2018, providing additional documentation and information on aspects including the proposed bank's infrastructure and governance in hopes of strengthening its case.[66] In March 2020, Square received conditional approval for a banking license from the U.S. Federal Deposit Insurance Corporation and in 2021 expected to launch a new direct subsidiary, Square Financial Services, offering small business loans and deposit products.[67]

Conclusion

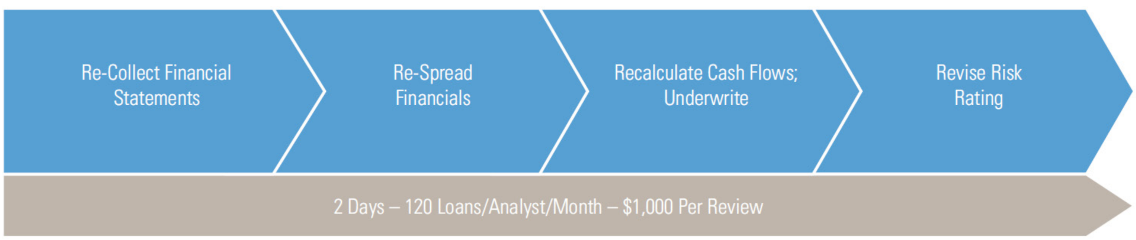
Digital technologies have disrupted conventional financial service models, simultaneously providing greater opportunities for SMEs to secure funding and raising alarms among some regulators and conventional banking institutions. Fintechs offer tremendous advantages when it comes to building financial cloud platforms, risk control systems, and digital distribution channels designed to reach the most people. However, banks continue to bring strengths, such as large balance sheets and critically needed industry experience in fraud and data protection, to the table.

Exhibit 1: Typical Underwriting Process



Source: Greg Ulankiewicz, "Gimme Credit: Faster, Simpler, Safer Credit for Main Street America," Raddon Research Insights, August 2018, 31.

Exhibit 2: Typical Loan Review Process



Source: Greg Ulankiewicz, "Gimme Credit: Faster, Simpler, Safer Credit for Main Street America," Raddon Research Insights, August 2018, 31.

Exhibit 3: Sample Earnings Implications for a Loan Using Typical Underwriting and Review Processes

Example #1

Assumptions	
Desired ROE	15.00%
Capital Requirement	10.0%
Loan Term	60 Months
—	
Loan Interest Rate	5.00%
Loan Loss Allocation	0.50%
Transfer Price[1]	2.84%
Discount Rate	4.06%
Current Process Costs	
Origination Costs	
Underwriting	$5,000
Servicing Costs	
Loan Review (Per Time)	$1,000
Breakeven Balance	
Annual Review	$861,505
Semiannual Review	$1,274,761
Quarterly Review	$2,101,316

[1] 5 year Treasury Rate as of April 2018

Example #2

Assumptions	
Desired ROE	15.00%
Capital Requirement	10.0%
Loan Term	60 Months
Loan Amount	$75,000
—	
Loan Loss Allocation	0.50%
Transfer Price[1]	2.84%
Discount Rate	4.06%
Current Process Costs	
Origination Costs	
Underwriting	$5,000
Servicing Costs	
Loan Review (Per Time)	$1,000
Breakeven Interest Rate	
Annual Review	9.49%
Semiannual Review	11.74%
Quarterly Review	16.05%

[1] 5 year Treasury Rate as of April 2018

Example #3

Assumptions	
—	
Capital Requirement	10.0%
Loan Term	60 Months
Loan Amount	$75,000
Loan Interest Rate	5.00%
Loan Loss Allocation	0.50%
Transfer Price[1]	2.84%
Discount Rate	4.06%
Current Process Costs	
Origination Costs	
Underwriting	$5,000
Servicing Costs	
Loan Review (Per Time)	$1,000
Return on Equity	
Annual Review	–30.89%
Semiannual Review	–56.40%
Quarterly Review	–110.95%

[1] 5 year Treasury Rate as of April 2018

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